

ENVS193_Final

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Set Up

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(gt)
library(DHARMA)
```

This is DHARMA 0.5.0. For overview type '?DHARMA'. For recent changes, type `news(package = 'DHARMA')`.

Note that the default setting in `simulateResiduals` was changed to conditional simulations since version 0.5.0. For more details, see `?simulateResiduals`.

```
 nests <- read_csv("nest_data_final.csv")
```

```
Rows: 299 Columns: 10
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (3): alive, bird_species, ant_species
```

```
dbl (7): height_cm, diameter_mm, case_control, case, canopy_width, canopy_le...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Problem 1: Research writing

a) Transparent statistical methods

Part 1: The co-worker likely used a logistic regression because the response variable is the probability of American Avocet microhabitat use (used vs. not used) and the predictor variable is distance to the water edge, which is continuous.

Part 2: The co-worker likely used a chi-square test of independence because both bird species and habitat type are categorical variables, and the goal is to determine whether the distribution of bird species differs among habitat types.

b) Suggestions For Rewriting

Part 1: American Avocets were more likely to use some areas than others depending on their distance from the water's edge, suggesting that proximity to water influences microhabitat selection.

Part 2: The three bird species did not use habitats in the same way. Habitat use differed among species, indicating that each species may prefer different habitat types within the San Francisco Bay.

c) Further Analysis

One additional variable that could influence the probability of American Avocet microhabitat use is water depth (cm), measured at each survey location using a meter stick or depth gauge. Water depth may affect habitat use because American Avocets forage by sweeping their bills through shallow water, so they may prefer areas with depths that maximize feeding efficiency.

Problem 2 Data analysis

a) Clean and Wrangle

```
# Create a new cleaned dataset containing only the variables needed for the nest occurrence analysis
nests_clean <- nests |>
  # Keep only the response variable, tree height, and ant species columns
  select(case_control, height_cm, ant_species) |>
  # Keep only trees occupied by the three focal ant species
  filter(ant_species %in% c("AB", "BBR", "RRB")) |>
  # Recode and reorder ant species information
  mutate(
    # Convert the ant species codes into a factor and set the desired order
    ant_species = factor(
      ant_species,
      levels = c("BBR", "AB", "RRB"),
      labels = c(
        "C. sjostedti",
        "C. mimosae",
        "C. nigriceps"
      )
    ),
    # Create a new column containing the full scientific name of each ant species
    ant_species_full = case_when(
      ant_species == "C. sjostedti" ~ "Crematogaster sjostedti",
      ant_species == "C. mimosae" ~ "Crematogaster mimosae",
      ant_species == "C. nigriceps" ~ "Crematogaster nigriceps"
    )
  )
# Display the structure of the cleaned dataset to verify variable types and dimensions
str(nests_clean)
```

tibble [270 x 4] (S3: tbl_df/tbl/data.frame)

```
$ case_control      : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
$ height_cm        : num [1:270] 392 310 513 154 324 ...
$ ant_species       : Factor w/ 3 levels "C. sjostedti",...: 3 3 3 3 2 3 1 3 2 1 ...
$ ant_species_full: chr [1:270] "Crematogaster nigriceps" "Crematogaster nigriceps" "Crematogaster nigriceps" ...
```

```
# Display 10 randomly selected rows from the cleaned dataset
nests_clean |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
  case_control height_cm ant_species ant_species_full
      <dbl>      <dbl> <fct>          <chr>
1         0        187 C. nigriceps  Crematogaster nigriceps
2         0        243 C. nigriceps  Crematogaster nigriceps
3         1       406. C. nigriceps  Crematogaster nigriceps
4         0        142 C. nigriceps  Crematogaster nigriceps
5         0       132. C. mimosae    Crematogaster mimosae
6         1       212. C. nigriceps  Crematogaster nigriceps
7         1       392. C. nigriceps  Crematogaster nigriceps
8         0       122. C. nigriceps  Crematogaster nigriceps
9         1       267 C. nigriceps  Crematogaster nigriceps
10        0       114 C. nigriceps  Crematogaster nigriceps
```

b) Response variable

The response variable is nest occurrence, where a value of 1 indicates that a bird nest was present in a tree and a value of 0 indicates that no bird nest was present. Biologically, this variable represents whether a tree was selected and used as a nesting site by birds.

c) Purpose of Study

Tree height:

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence may increase because taller trees can provide better nesting locations and may be more visible or accessible to birds while also reducing the risk of some predators. This information is discussed in the Methods section, “Nest Site Selection” subsection, where tree characteristics were measured, and is supported by the Results section, where tree height was included as a predictor of nest occurrence.

Acacia ant species:

Acacia ant species is a categorical variable with three levels: *C. sjostedti*, *C. mimosae*, and *C. nigriceps*. The probability of nest occurrence may differ among ant species because birds preferentially nest in trees occupied by more aggressive ant species that defend trees against browsing mammals and potential nest predators. This information is described in the Introduction (final paragraph) and supported in the Discussion section, where the authors explain that birds selected trees inhabited by the more aggressive ant species.

d) Table of Models

```
# Create a table that summarizes the two candidate logistic regression models
model_table <- tibble(
  # Give each model a descriptive name
  `Model Name` = c("Additive Model", "Interactive Model"),
  # Identify the response variable for both models
  `Response Variable` = c(
    "Nest occurrence",
    "Nest occurrence"
  ),
  # Describe the predictors included in each model
  `Predictors` = c(
    "Tree height + acacia ant species",
    "Tree height + acacia ant species + tree height x acacia ant species"
  ),
  # State whether each model includes an interaction term
  `Interaction Term` = c(
    "No",
    "Yes"
  )
)
# Display the model table
model_table |>
  gt() |>
  tab_header(
    title = "Candidate Logistic Regression Models"
  )
```

e) Run the models

Candidate Logistic Regression Models

Model Name	Response Variable	Predictors
Additive Model	Nest occurrence	Tree height + acacia ant species
Interactive Model	Nest occurrence	Tree height + acacia ant species + tree height x acacia ant species

```
# Run the additive logistic regression model
additive_model <- glm(
  case_control ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)
# Run the interaction logistic regression model
interaction_model <- glm(
  case_control ~ height_cm * ant_species,
  data = nests_clean,
  family = binomial
)
```

f) Select the best model

```
AIC(additive_model, interaction_model)
```

```
              df      AIC
additive_model  4 258.7430
interaction_model 6 237.8042
```

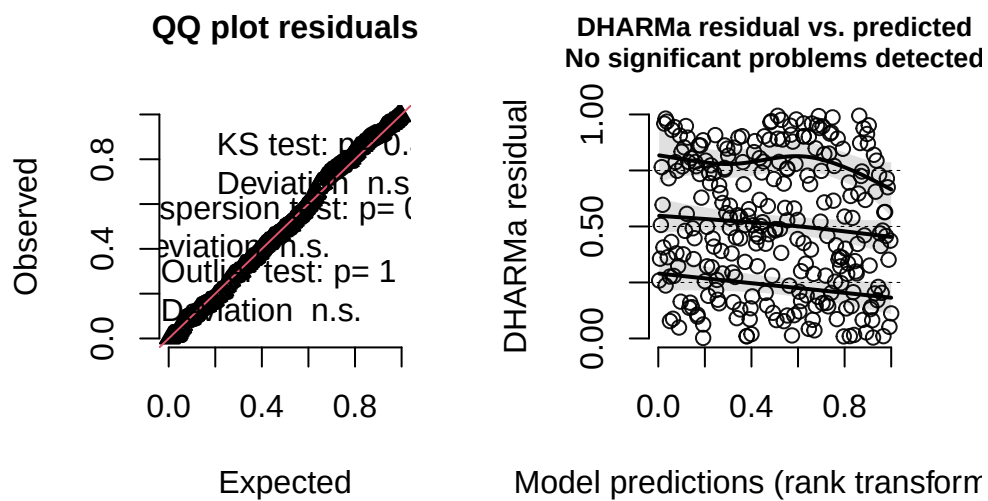
The best model as determined by Akaike's Information Criterion (AIC) was the logistic regression model predicting nest occurrence from tree height, acacia ant species, and the interaction between tree height and acacia ant species. This model had a lower AIC value (237.804) than the model containing only the additive effects of tree height and acacia ant species (258.743), indicating stronger support for the interaction model.

g) Check the diagnostics

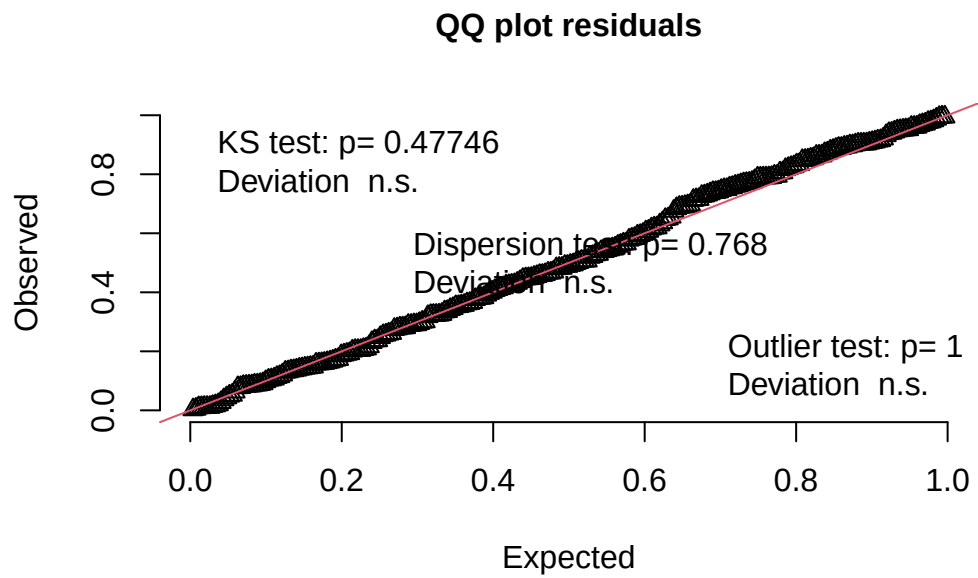
```
# Simulate residuals from the selected model
interaction_residuals <- simulateResiduals(
  fittedModel = interaction_model
)

# Diagnostic plots
plot(interaction_residuals)
```

DHARMa residual



```
# Test for uniformity
testUniformity(interaction_residuals)
```

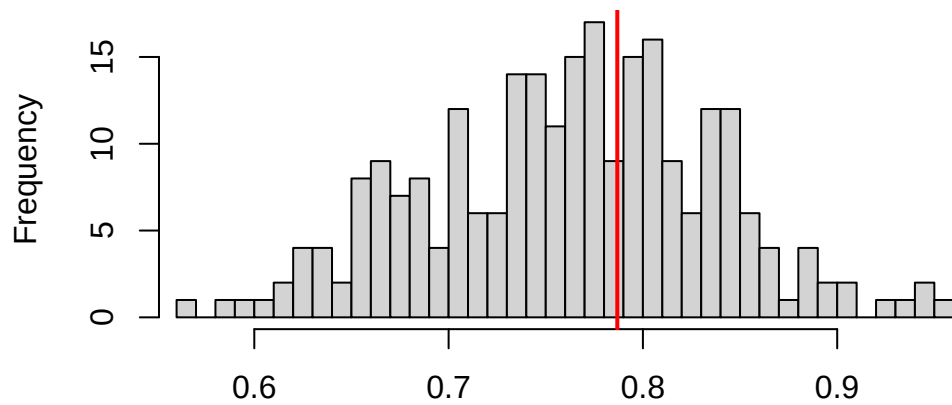


Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals  
D = 0.051246, p-value = 0.4775  
alternative hypothesis: two-sided
```

```
# Test for dispersion  
testDispersion(interaction_residuals)
```


**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**

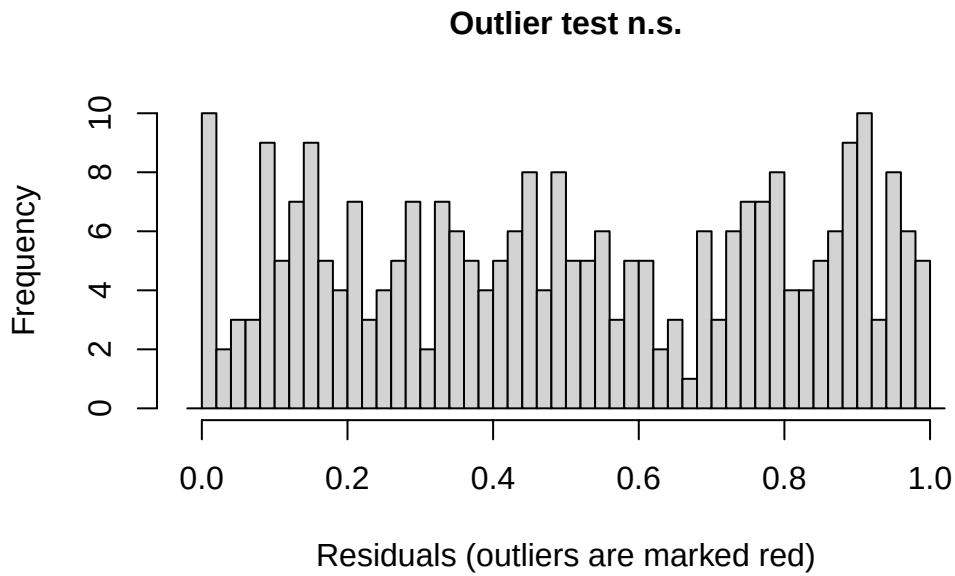


Simulated values, red line = fitted model. p-value (two.sided) = 0.768

DHARMA nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput
dispersion = 1.0325, p-value = 0.768
alternative hypothesis: two.sided
```

```
# Test for outliers
testOutliers(interaction_residuals)
```



DHARMa bootstrapped outlier test

```
data: interaction_residuals
outliers at both margin(s) = 0, observations = 270, p-value = 1
alternative hypothesis: two.sided
percent confidence interval:
 0 0
sample estimates:
outlier frequency (expected: 7.40740740740741e-05 )
0
```

h) Visualize the Model Predictions

Predictions

```
# Create a sequence of tree heights for each ant species
prediction_data <- expand.grid(
  height_cm = seq(
    min(nests_clean$height_cm),
    max(nests_clean$height_cm),
    length.out = 100
```

```

),
  ant_species = levels(nests_clean$ant_species)
)

# Generate model predictions
predictions <- predict(
  interaction_model,
  newdata = prediction_data,
  type = "link",
  se.fit = TRUE
)

# Calculate predicted probabilities and confidence intervals
prediction_data <- prediction_data |>
  mutate(
    predicted_probability = plogis(predictions$fit),
    lower_ci = plogis(predictions$fit - 1.96 * predictions$se.fit),
    upper_ci = plogis(predictions$fit + 1.96 * predictions$se.fit)
  )

```

Graph

```

# Create a plot showing the observed nest occurrence data and the predicted probabilities from the model
ggplot() +

# Add the underlying data as jittered points
  geom_jitter(
    data = nests_clean,
    aes(
      x = height_cm,
      y = case_control
    ),
    width = 0,
    height = 0.04,
    alpha = 0.25,
    size = 1.6
  ) +

# Add shaded 95% confidence intervals around the model predictions
  geom_ribbon(
    data = prediction_data,

```

```

aes(
  x = height_cm,
  ymin = lower_ci,
  ymax = upper_ci,
  fill = ant_species
),
alpha = 0.25
) +

# Add the predicted probability lines from the interaction model
geom_line(
  data = prediction_data,
  aes(
    x = height_cm,
    y = predicted_probability,
    color = ant_species
  ),
  linewidth = 1.2
) +

# Create separate panels for each ant species
facet_wrap(~ ant_species) +

# Set y-axis limits and tick marks to represent probabilities
scale_y_continuous(
  breaks = seq(0, 1, 0.25)
) +
coord_cartesian(
  ylim = c(0, 1)
) +

# Add informative axis labels
labs(
  x = "Tree height (cm)",
  y = "Predicted probability of bird nest occurrence"
) +

# Use a clean, non-default theme
theme_classic() +

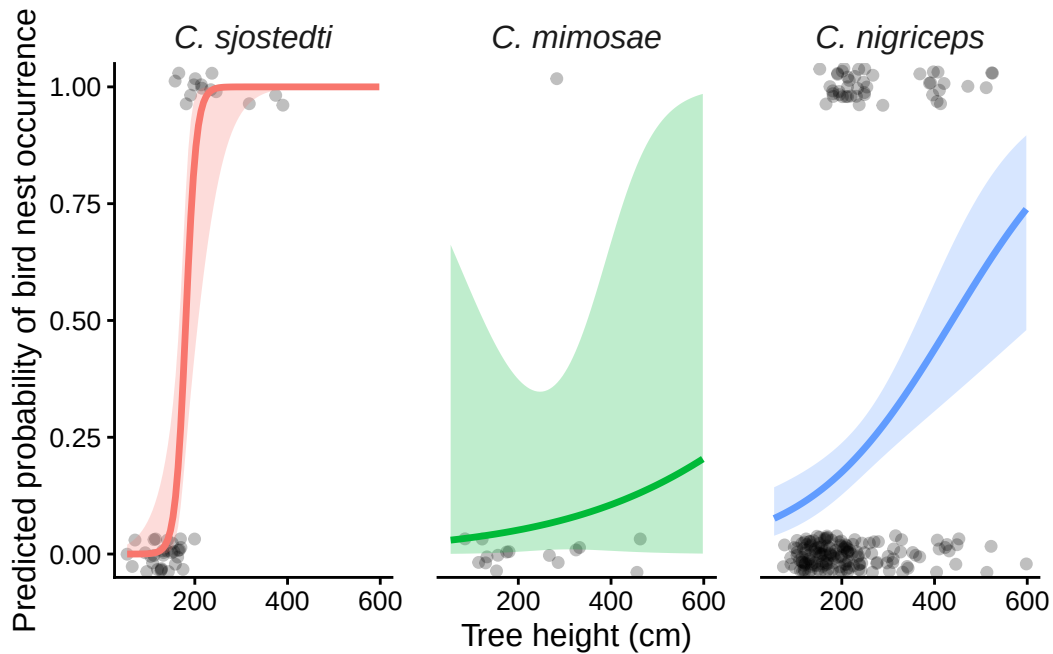
# Remove the legend and improve readability
theme(

```

```

legend.position = "none",
strip.background = element_blank(),
strip.text = element_text(
  size = 12,
  face = "italic"
),
axis.title = element_text(size = 12),
axis.text = element_text(size = 10),
panel.spacing = unit(1.2, "lines")
)

```



i) Write a Caption

Figure 1. Predicted probability of bird nest occurrence by tree height and acacia ant species. Points represent individual trees, where 0 indicates no bird nest was present and 1 indicates a bird nest was present. Colored lines show the predicted probability of nest occurrence from the selected logistic regression model, and shaded ribbons represent 95% confidence intervals around the predictions. Separate panels display predictions for trees occupied by *Crematogaster sjostedti*, *Crematogaster mimosae*, and *Crematogaster nigriceps*. Data source: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for:

Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo.
<https://doi.org/10.5281/zenodo.8373322>

j) Calculate Model Predictions

```
# Create a new data frame containing a tree height of 600 cm for each of the three acacia ant species
prediction_600 <- tibble(
  height_cm = 600,
  ant_species = factor(
    c(
      "C. sjostedti",
      "C. mimosae",
      "C. nigriceps"
    ),
    levels = levels(nests_clean$ant_species)
  )
)

# Calculate predicted probabilities using the selected interaction model
prediction_600 <- prediction_600 |>
  mutate(
    predicted_probability = predict(
      interaction_model,
      newdata = prediction_600,
      type = "response"
    )
  )

# Display the predicted probabilities
prediction_600
```

```
# A tibble: 3 x 3
  height_cm ant_species predicted_probability
    <dbl> <fct>          <dbl>
1     600 C. sjostedti          1
2     600 C. mimosae         0.205
3     600 C. nigriceps         0.741
```